Sahaj Bhadja

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Summary

Embedded-leaning software engineer with 2+ years delivering **C++/Python systems**, **automated test workflows**, and **Linux-ready deployment pipelines** with strong root-cause debugging, SDLC discipline, and production reliability focus.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Implemented **fault-tolerant FastAPI services** for 5K+ monthly background jobs with explicit state transitions, validation gates, and recovery paths, improving software stability in asynchronous, production-critical execution environments.
- Performed root-cause analysis on data-path bottlenecks and concurrency issues, reducing latency from **1.2s to 350ms** for 500+ users through indexing, query tuning, and safer interface-level integration patterns.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Built and tested **high-throughput service components** handling 10K+ daily requests, integrating OCR/model pipelines and enforcing robust input/output validation across parallel processing stages and API boundaries.
- Strengthened release quality with **CI/CD gates**, regression checks, and deterministic test paths, reducing turnaround by **60%** while preserving reliability under fast-changing requirements and iterative rollout schedules.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Developed **Python/Java microservice workflows** and integration interfaces on AWS Glue/S3, improving response times by **30%** while ensuring consistent behavior across distributed, multi-environment production software deployments.
- Maintained **20+ automated unit/integration suites** with Jenkins, PyTest, and Mockito, increasing pipeline efficiency by **68%** and reducing technical debt by **20%** through continuous defect prevention.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Built **REST/WebSocket interface modules** supporting eight workflows and 2M+ synchronized records, improving communication reliability between software components and reducing state inconsistency during real-time operational updates.
- Automated Docker and Jenkins build-test-release loops, cutting deployment time by **50%** and improving repeatability of software changes across Linux-based CI environments and collaborative engineering workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Designed **Python evaluation pipelines** over 100K+ trajectory logs, producing reproducible diagnostics and algorithm benchmarks that supported rigorous requirement validation and rapid iteration on system-level behavior assumptions.
- Documented technical findings and remediation strategies for multidisciplinary stakeholders, improving design review quality and accelerating decisions on robust implementation tradeoffs for complex, performance-sensitive software components.

Skills

Languages: **C++**, **Python**, **Java**, Bash scripting, SQL
Embedded/Systems: OOP design, interface/component design, root-cause analysis, performance tuning, Linux-based development
Quality & Process: SDLC, SCM workflows, Jenkins CI, unit/integration testing, code/design reviews, Agile Scrum
Infrastructure: Docker, Kubernetes, Git, AWS (S3/Glue), Azure SQL, observability-driven debugging

Projects

CPU Scheduling Simulator | *React, JavaScript, OS Algorithms*

- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling with Gantt visualization, reducing average turnaround by **25%** and improving debugging efficiency by **20%** across 1K+ simulated process scenarios.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built a **cross-platform Java/TypeScript toolchain** for schema-aware SQL execution across 10+ databases, improving correctness checks, workflow automation, and robustness of data operations in complex multi-environment setups.

PiXMod | *Python, Next.js, MongoDB, Transformers*

- Developed an **AI-driven image processing pipeline** with segmentation and retrieval modules, processing 1K+ assets while optimizing algorithm performance and improving repeatability of output quality under iterative parameter tuning.